Recap: Lecture 20

In Lecture 20, we

- Introduced BVPs (linear, 2nd-order)

$$f_2(x)y''(x) + f_1(x)y'(x) + f_0(x)y(x) = g(x), \qquad \begin{cases} a_1 y'(x_1) + b_1 y(x_1) & = c_1 \\ a_2 y'(x_2) + b_2\, y(x_2) & = c_2 \end{cases}$$

- A Dirichlet BC fixes the value of y at a boundary point and a Neumann BC fixes y'
- BVP solutions are solutions to homogeneous BVP ($g = c_1 = c_2 = 0$; form a solution space) + particular solution

Lecture 21

In Lecture 21, we will:

1. Introduce and classify PDEs
2. Introduce initial–boundary value problems
3. Introduce the 1D heat equation

The Laplacian (Vector Calculus revision)

The Laplacian is a linear transformation (differential operator) that appears in many common PDEs. It is denoted by ∇^2 and its representation changes depending on the coordinate system:

Δ in physics

The Laplacian (Vector Calculus revision)

The Laplacian is a linear transformation (differential operator) that appears in many common PDEs. It is denoted by ∇^2 and its representation changes depending on the coordinate system:

1. 2D Cartesian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

The Laplacian (Vector Calculus revision)

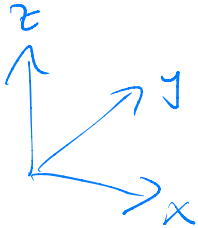
The Laplacian is a linear transformation (differential operator) that appears in many common PDEs. It is denoted by ∇^2 and its representation changes depending on the coordinate system:

1. 2D Cartesian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

2. 3D Cartesian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$



The Laplacian (Vector Calculus revision)

The Laplacian is a linear transformation (differential operator) that appears in many common PDEs. It is denoted by ∇^2 and its representation changes depending on the coordinate system:

$\nabla^2 = d^2/dx^2$ in 1D

1. 2D Cartesian

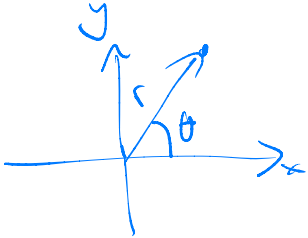
$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

2. 3D Cartesian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

3. Cylindrical

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}$$



PDEs and Classification

If there are multiple independent variables then the differential operators involve partial derivatives.

PDEs and Classification

If there are multiple independent variables then the differential operators involve partial derivatives.

1. The *order* of a PDE is the order of the highest derivative.

PDEs and Classification

If there are multiple independent variables then the differential operators involve partial derivatives.

1. The *order* of a PDE is the order of the highest derivative.

2. A PDE is *linear* if it is a linear combination of partial derivatives of the unknown function, whose coefficients may only be functions of the independent variables, which equals a function of the independent variables. If it does not meet this criteria it is *non-linear*.

PDEs and Classification

If there are multiple independent variables then the differential operators involve partial derivatives.

1. The *order* of a PDE is the order of the highest derivative.

2. A PDE is *linear* if it is a linear combination of partial derivatives of the unknown function, whose coefficients may only be functions of the independent variables, which equals a function of the independent variables. If it does not meet this criteria it is *non-linear*.

3. A PDE is *homogeneous* if each term contains the unknown function. It is *inhomogeneous* if there is a term which contains only the independent variables.

Partial differential equations (PDEs)

Examples of some PDEs include:

Partial differential equations (PDEs)

Examples of some PDEs include:

- Laplace's equation

$$\nabla^2 u = 0$$

Partial differential equations (PDEs)

Examples of some PDEs include:

- Laplace's equation

$$\nabla^2 u = 0$$

- Heat/diffusion equation

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

Partial differential equations (PDEs)

Examples of some PDEs include:

- Laplace's equation

$$\nabla^2 u = 0$$

- Heat/diffusion equation

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

- Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

Partial differential equations (PDEs)

Examples of some PDEs include:

- Laplace's equation

$$\nabla^2 u = 0$$

- Heat/diffusion equation

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

- Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

- Schrödinger's equation

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$$

- Liouville's equation

$$\nabla^2 u + \mathrm{e}^{\lambda u} = 0$$

- Burgers' equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

Exercise 5.2: Classify the following PDEs

(a) $\nabla^2 f = x^2 y$, (b) $\dfrac{\partial^2 u}{\partial x \partial y} + \sin x \dfrac{\partial u}{\partial x} = 0$, (c) $u \dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} = \mathrm{e}^x$, (d) $\nabla^2 u = -\mathrm{e}^{\lambda u}$.

Exercise 5.2: Classify the following PDEs

(a) $\nabla^2 f = x^2 y$, (b) $\dfrac{\partial^2 u}{\partial x \partial y} + \sin x \dfrac{\partial u}{\partial x} = 0$, (c) $u\dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} = \mathrm{e}^x$, (d) $\nabla^2 u = -\mathrm{e}^{\lambda u}$.

(a) 2nd order, linear and inhomogeneous
(b) 2nd order, linear and homogeneous
(c) 1st order, non-linear and inhomogeneous
(d) 2nd order, non-linear and homogeneous

Initial and boundary conditions

- General solutions of PDEs can contain *unknown functions*
  - General solutions of ODEs have unknown constants, which we try to specify with initial or boundary conditions

Initial and boundary conditions

- General solutions of PDEs can contain *unknown functions*
  - General solutions of ODEs have unknown constants, which we try to specify with initial or boundary conditions

Initial and boundary conditions

- General solutions of PDEs can contain *unknown functions*
  - General solutions of ODEs have unknown constants, which we try to specify with initial or boundary conditions
- If the PDE is first order in time, we usually constrain the unknown function to be some specified function, $u(\boldsymbol{x}, t)$ at the initial time, i.e., $u(\boldsymbol{x}, 0) = f(\boldsymbol{x})$

$\underline{x} = x$ in 1D

$= (x, y)$ in 2D

$= (x, y, z)$ in 3D

Initial and boundary conditions

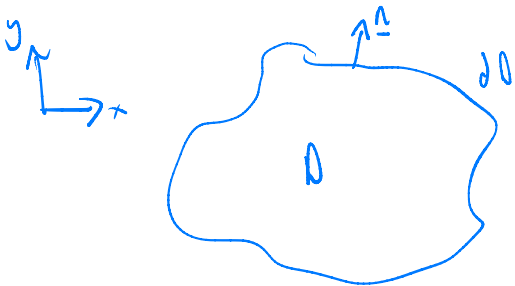
- General solutions of PDEs can contain *unknown functions*
  - General solutions of ODEs have unknown constants, which we try to specify with initial or boundary conditions
- If the PDE is first order in time, we usually constrain the unknown function to be some specified function, $u(\boldsymbol{x}, t)$ at the initial time, i.e., $u(\boldsymbol{x}, 0) = f(\boldsymbol{x})$
- If it is second order in time, we impose an additional constraint on the *time* derivative of the unknown function, i.e., $\partial_t u(\boldsymbol{x}, 0) = g(\underline{x})$

Initial and boundary conditions

- Boundary conditions are constraints imposed at specified locations in space for all times after the initial condition.

Initial and boundary conditions

- Boundary conditions are constraints imposed at specified locations in space for all times after the initial condition.
- Let D be a bounded spatial domain, then boundary conditions are imposed on the domain's boundary, ∂D.



Initial and boundary conditions

- Boundary conditions are constraints imposed at specified locations in space for all times after the initial condition.
- Let D be a bounded spatial domain, then boundary conditions are imposed on the domain's boundary, ∂D.
- Two most common boundary conditions are:
  1. *Dirichlet BCs* fix the value of the unknown function on the boundary, i.e., $u(\boldsymbol{x}, t) = f(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$.

Initial and boundary conditions

- Boundary conditions are constraints imposed at specified locations in space for all times after the initial condition.
- Let D be a bounded spatial domain, then boundary conditions are imposed on the domain's boundary, ∂D.
- Two most common boundary conditions are:
  1. *Dirichlet BCs* fix the value of the unknown function on the boundary, i.e., $u(\boldsymbol{x}, t) = f(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$.
  2. *Neumann BCs* fix the value of the normal derivative at the boundary, i.e., $\partial_{\boldsymbol{n}} u(\boldsymbol{x}, t) = g(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$ and $\boldsymbol{n}$ is the boundary's outwards unit normal.

Initial and boundary conditions

- Boundary conditions are constraints imposed at specified locations in space for all times after the initial condition.
- Let D be a bounded spatial domain, then boundary conditions are imposed on the domain's boundary, ∂D.
- Two most common boundary conditions are:
  1. *Dirichlet BCs* fix the value of the unknown function on the boundary, i.e., $u(\boldsymbol{x}, t) = f(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$.
  2. *Neumann BCs* fix the value of the normal derivative at the boundary, i.e., $\partial_{\boldsymbol{n}} u(\boldsymbol{x}, t) = g(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$ and $\boldsymbol{n}$ is the boundary's outwards unit normal.

A PDE with ICs and BCs is called an *initial-boundary value problem* (IBVP)

What will we solve?

1. The two-dimensional Laplace equation in Cartesian and polar coordinates

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{and} \qquad \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

*We will also solve some inhomogeneous examples

What will we solve?

1. The two-dimensional Laplace equation in Cartesian and polar coordinates

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{and} \qquad \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

2. The heat/diffusion equation in one and two-dimensions, in Cartesian and axisymmetric polar coordinates

$$\frac{\partial u}{\partial t} = \alpha\frac{\partial^2 u}{\partial x^2}, \qquad \frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \qquad \text{and} \qquad \frac{\partial u}{\partial t} = \frac{\alpha}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right).$$

*We will also solve some inhomogeneous examples

What will we solve?

1. The two-dimensional Laplace equation in Cartesian and polar coordinates

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{and} \qquad \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

2. The heat/diffusion equation in one and two-dimensions, in Cartesian and axisymmetric polar coordinates

$$\frac{\partial u}{\partial t} = \alpha\frac{\partial^2 u}{\partial x^2}, \qquad \frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \qquad \text{and} \qquad \frac{\partial u}{\partial t} = \frac{\alpha}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right).$$

3. The wave equation in one and two-dimensions, in Cartesian and axisymmetric polar coordinates

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}, \qquad \frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \qquad \text{and} \qquad \frac{\partial^2 u}{\partial t^2} = \frac{c^2}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)$$

*We will also solve some inhomogeneous examples

Classification of second-order PDEs

Suppose the PDE is of the form

$$A\,u_{xx} + B\,u_{xy} + C\,u_{yy} + \text{lower-order terms} = 0$$

Classification of second-order PDEs

Suppose the PDE is of the form

$$A\,u_{xx} + B\,u_{xy} + C\,u_{yy} + \text{lower-order terms} = 0$$

The PDE is:

Classification of second-order PDEs

Suppose the PDE is of the form

$$A\,u_{xx} + B\,u_{xy} + C\,u_{yy} + \text{lower-order terms} = 0$$

The PDE is:

- *Elliptic* if $B^2 - 4\,A\,C < 0$

Classification of second-order PDEs

Suppose the PDE is of the form

$$A\,u_{xx} + B\,u_{xy} + C\,u_{yy} + \text{lower-order terms} = 0$$

The PDE is:

- *Elliptic* if $B^2 - 4\,A\,C < 0$
- *Parabolic* if $B^2 - 4\,A\,C = 0$

Classification of second-order PDEs

Hyperbolic → wave eqn in 1D

$c^2 u_{xx} - u_{tt} = 0 \;\|\; A = c^2,\ C = -1,\ B = 0$

$B^2 - 4AC = c^2 > 0$

Suppose the PDE is of the form

$$A\,u_{xx} + \overset{2}{B}\,u_{xy} + C\,u_{yy} + \text{lower-order terms} = 0$$

The PDE is:

- *Elliptic* if $B^2 - \cancel{4}A\,C < 0$
- *Parabolic* if $B^2 - \cancel{4}A\,C = 0$
- *Hyperbolic* if $B^2 - \cancel{4}A\,C > 0$

elliptic → laplace eqn

$u_{xx} + u_{yy} = 0 \;\|\; A = C = 1,\ B = 0$

$\therefore B^2 - 4AC = -4 < 0$

Parabolic → heat eqn in 1D

$\alpha u_{xx} - u_t = 0 \;\|\; A = \alpha,\ C = 0,\ B = 0$

$\therefore B^2 - 4AC = 0$

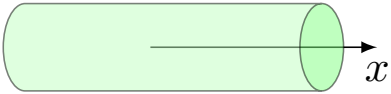
Case study 7: Heat conduction in a quartz rod

- **Quartz** is unique due to its special mechanical, electrical, thermal, chemical and optical properties.
- It has an exceptional thermal stability. For example, a piece of quartz will not break if heated to a temperature of 1100°C, then suddenly cooled down to 20°C.
- **WHY CARE? Well this is super important if you have ambitions of becoming 007 since your job will require you to run through burning buildings and cars – all in a day's work!**

So .. this is why Bond only wears Quartz watches.

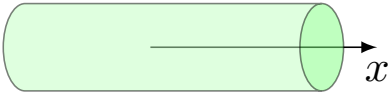
The 1D heat equation (below derivation won't be examined)

- Consider a straight rod of uniform cross section and homogeneous matrial (quartz)



The 1D heat equation (below derivation won't be examined)

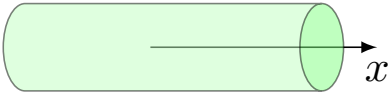
- Consider a straight rod of uniform cross section and homogeneous matrial (quartz)



- Let the temperature along the rod be $u(x, t)$, where x is axial location and t is time

The 1D heat equation (below derivation won't be examined)

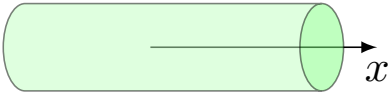
- Consider a straight rod of uniform cross section and homogeneous matrial (quartz)



- Let the temperature along the rod be $u(x,t)$, where x is axial location and t is time
- Fourier's law: $\text{Heat flow} = -\kappa \dfrac{\partial u}{\partial x}$ where $\kappa = \text{thermal conductivity}$

The 1D heat equation (below derivation won't be examined)

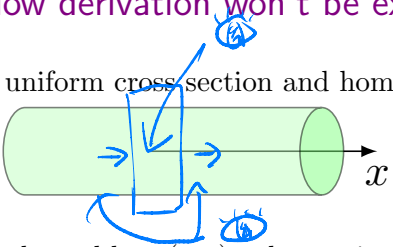
- Consider a straight rod of uniform cross section and homogeneous matrial (quartz)



- Let the temperature along the rod be $u(x,t)$, where x is axial location and t is time
- Fourier's law: Heat flow $= -\kappa \dfrac{\partial u}{\partial x}$ where $\kappa =$ thermal conductivity
- Rate of change in heat $= c\,\rho \dfrac{\partial u}{\partial t}$ where $c =$ specific heat capacity, $\rho =$ density

The 1D heat equation (below derivation won't be examined)

- Consider a straight rod of uniform cross section and homogeneous matrial (quartz)



- Let the temperature along the rod be $u(x,t)$, where x is axial location and t is time

- Fourier's law: Heat flow $= -\kappa \dfrac{\partial u}{\partial x}$ where $\kappa =$ thermal conductivity

- Rate of change in heat $= c\,\rho \dfrac{\partial u}{\partial t}$ where $c =$ specific heat capacity, $\rho =$ density

- Conservation law: $c\,\rho \dfrac{\partial u}{\partial t} = -\dfrac{\partial}{\partial x}\left(-\kappa \dfrac{\partial u}{\partial x}\right) \quad \Rightarrow \quad \dfrac{\partial u}{\partial t} = \alpha \dfrac{\partial^2 u}{\partial x^2}$

  where $\alpha = \kappa / c\,\rho =$ thermal diffusivity

Thermal diffusivity

Material	α [$\mathrm{cm^2\,s^{-1}}$]
Gold	1.27
Copper	1.11
Silicon	8.8×10^{-1}
Aluminium	8.4×10^{-1}
Iron	2.3×10^{-1}
Air	1.9×10^{-1}
Steel	1.2×10^{-1}
Quartz	1.4×10^{-2}
Brick	3.8×10^{-3}
Water	1.4×10^{-3}
Wood	8.2×10^{-4}